L4- Problem Solving

Tuesday, August 11, 2020 10:44 AM

Class Activity1:

Find two elements in the list of n numbers that sum up to a given value (**val**). The list is not sorted.

The function shall return indices of the two elements that sum up to the given value or return -1,-1 in case of failure.

Let the list be of five elements a_1, a_2, a_3, a_4, a_5
Find sum of all pairs.
Compare each pair with the given value (val)

$(a_1,a_2), (a_1,a_3), (a_1,a_4), (a_1,a_5),$
$(a_2,a_3), (a_2,a_4), (a_2,a_5),$
$(a_3,a_4), (a_3,a_5),$
(a4,a5),

TwoNumbers(Arr, val)
// The input list, Arr, is un-sorted

```
1. for i = 1 to Arr.length-1 {
2.     for j = i+1 to Arr.length {
3.         if Arr[i] + Arr[j] == val
4.                 return i , j
5.     }
6. }
7. return -1, -1
```

TwoNumbers(Arr, sum)
// The input list, Arr, is un-sorted

```
                                              Operations    Worst Case Frequency
1. for i = 1 to Arr.length-1 {                  c1            n
2.     for j = i+1 to Arr.length{               c2            n(n+1)/2 - 1
3.         if Arr[i] + Arr[j] == sum            c3            n(n-1)/2
4.             return i , j                     c4            1
5.     }
6. }
7. return -1 , -1                               c5            1
```

// either Line 4 or Line 7 will be executed, not Both

Let Arr.length be n.
Let n=5

If i == 1, then Line 2 executes 5 (n-times)
If i == 2, then Line 2 executes 4 (n-1-times)
If i == 3, then Line 2 executes 3 (n-2-times)
If i == 4, then Line 2 executes 2 (n-3-times)
If i == 5, then Line 2 executes 0 (n-n-times)

Line 2 executes a total of $2+3+4+5 = 14$

Line 2 executes a total of $2+3+\ldots+n-1+n$

Line 2 executes a total of $2+3+\ldots+n-1+n = (n+2)(n-1)/2 = (n*n - n +2n - 2)/2$

$$= (n*n +n -2)/2$$
$$= (n*n + n)/2 - 2/2$$
$$= n(n+1)/2 - 1$$

If i == 1, then Line 3 executes 4 (n-1 -times)

If i == 2, then Line 3 executes 3 (n-2-times)

If i == 3, then Line 3 executes 2 (2 -times)

If i == 4, then Line 3 executes 1 (1 -times)

Line 3 executes a total of $1+2+3+4 = 10$

Line 3 executes a total of $1+2+\ldots +n-2+n-1 = n(n-1)/2$

Worst case arises when the two elements are the last two elements of the list or no pair of elements add up to given value

$$\begin{aligned} T(n) &= c1(n)+(c2)(n(n+1)/2 - 1) + c3(n(n-1)/2) + c4 \\ &= c1(n) + (c2+c3)(n^2)/2 + (c2-c3)n/2 + c4 - c2 \\ &= c10(n^2) + c11(n) + c12 \\ &= O(n^2) \end{aligned}$$

Best case arises when the first two elements add up to the given value. In this case Lines 1 through 4 of the pseudo-code, given above, executes only one time.

$$\begin{aligned} T(n) &= c1+c2+c3+c4 \\ &= c10 \end{aligned}$$

Class Activity2:

Find two elements in the list of n numbers that sum up to a given value (**val**). **The list is sorted**.

The function shall return indices of the two elements that sum up to the given value or return -1,-1 in case of failure.

Let the list be as follows:

2 4 5 7 8 12 14 34 45 //val=48

i j

2+45 = 47 //given **val** will either be equal, greater or smaller than 47

4+45 = 49

4+34 = 38

5+34 = 39

7+34 = 41

8+34 = 42

12+34 = 46

14+34 = 48

TwoNumbers(Arr, sum)

// The input list, Arr, is sorted

1. let left = 1
2. let right = Arr.length

```
while (left < right)
        if (Arr[left] + Arr[right]) > sum
                right = right - 1
        else if (Arr[left] + Arr[right]) < sum
                left = left + 1
        else return left, right
}
return -1, -1
```